

RS Vorlesung 5

HA) 3) Var 1) $M \ddot{y} + k y + b \dot{y} = u$

$$a_0 = \frac{k}{M} \quad b_0 = \frac{1}{M}$$

$$\ddot{y} + \frac{b}{M} \dot{y} + \frac{k}{M} y = \frac{1}{M} u$$

$$a_1 = \frac{b}{M}$$

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ -\frac{k}{M} & -\frac{b}{M} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{1}{M} \end{pmatrix} u$$

$$y = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Var 2) $\left. \begin{array}{l} y = x_1 \\ \dot{y} = \dot{x}_1 = x_2 \\ \ddot{y} = \dot{x}_2 \end{array} \right\} \begin{array}{l} \dot{x}_2 + \frac{b}{M} x_2 + \frac{k}{M} x_1 = \frac{1}{M} u \\ \dot{x}_1 = x_2 \end{array} \right\}$

HA) 1) $\dot{x}_1 = 0 = x_{2,R}$

$$\Rightarrow x_{2,R} = 0$$

$$\dot{x}_3 = 0 = -\frac{R}{L} x_{3,R} + 0 + \frac{1}{L} u \Rightarrow x_{3,R} = \frac{1}{R} u_R$$

$$\dot{x}_2 = 0 = g - \frac{c u_R^2}{R^2 m x_1^2}$$

$$\Rightarrow x_{1,R} = \sqrt{\frac{c u_R^2}{R^2 m g}} = y_R$$

$$x_R = \begin{pmatrix} \frac{1}{R} \sqrt{\frac{c}{m g}} \\ 0 \\ \frac{1}{R} \end{pmatrix} u_R$$

$$u_R = R \sqrt{\frac{m g}{c}} y_R$$

$$x_R = \begin{pmatrix} 1 \\ 0 \\ \sqrt{\frac{m g}{c}} \end{pmatrix} y_R$$

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$$HA) 1) \quad \frac{\cancel{\partial x_R}}{\partial y_R} \frac{\partial f_1}{\partial x} \Big|_{AP} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\frac{\partial f_2}{\partial x} \Big|_{AP} = \begin{pmatrix} \frac{2c x_{3,R}^2}{m x_{1,R}^3} \\ 0 \\ -\frac{2c x_{3,R}^3}{m x_{1,R}^2} \end{pmatrix} \quad \begin{matrix} \frac{2c m g y_R^2}{x m y_R^3} \\ \\ -\frac{2 g \sqrt{m g} y_R}{\sqrt{c}} \end{matrix}$$

$$\frac{\partial f_3}{\partial \bar{x}} \Big|_{AP} = \begin{pmatrix} -\frac{2 \cdot 2c x_{2,R} x_{3,R}}{L x_{1,R}^3} \\ \frac{2c x_{3,R}}{L x_{1,R}^2} \\ -\frac{R}{L} + \frac{2c x_{2,R}}{L x_{1,R}^2} \end{pmatrix}$$

$$\frac{\partial f_1}{\partial u} \Big|_{AP} = \frac{\partial f_2}{\partial u} \Big|_{AP} = 0$$

$$\frac{\partial f_3}{\partial u} \Big|_{AP} = \frac{1}{L}$$

$$\frac{\partial f(\underline{x}, u)}{\partial \underline{x}} \Big|_{AP} = \begin{pmatrix} 0 & 1 & 0 \\ \frac{2g}{y_R} & 0 & -\sqrt{\frac{mg}{c}} 2g y_R \\ 0 & \frac{2\sqrt{mgc}}{L y_R} & -\frac{R}{L} \end{pmatrix} = \Delta A$$

$$\frac{\partial f(\underline{x}, u)}{\partial u} \Big|_{AP} = \begin{pmatrix} 0 \\ 0 \\ \frac{1}{L} \end{pmatrix} = \Delta \underline{b}$$

$$\frac{\partial g(\underline{x})}{\partial \underline{x}} \Big|_{AP}^T = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} = \Delta c^T$$

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HA) 2) $X_3 = \frac{1}{s+3} X_2 \quad Y = X_3 + 4U$

$$X_2 = \frac{1}{s-2} X_1$$

$$X_1 = \frac{1}{s+1} U$$

$$X_2 = (s+3)X_3 \Rightarrow X_2 = \dot{X}_3 + 3X_3$$

$$X_1 = (s-2)X_2 \Rightarrow X_1 = \dot{X}_2 - 2X_2$$

$$U = (s+1)X_1 \Rightarrow U = \dot{X}_1 + X_1$$

$$\underline{\dot{X}} = \begin{pmatrix} -1 & 0 & 0 \\ 1 & 2 & 0 \\ 0 & 1 & -3 \end{pmatrix} \underline{X} + \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} U$$

$$Y = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix} \underline{X} + 4U$$

HA) 4) $i_{C1} = \dot{U}_{C1} C \Rightarrow \dot{U}_{C1} = \frac{1}{C} i_{C1}$

$$U = U_{C1} + \dot{i}_L L$$

$$i_{C2} = \dot{U}_{C2} C \Rightarrow \dot{U}_{C2} = \frac{1}{C} i_{C2}$$

$$i_L = i_{C1} + i_R \quad i_R = i_{C2}$$

$$U_L = \dot{i}_L L \Rightarrow \dot{i}_L = \frac{1}{L} U_L$$

$$U_{C2} = -R i_R + \dot{i}_L L$$

$$U_{C2} = +R(i_L - C \dot{U}_{C1}) + \frac{1}{L} U - U_{C1}$$

$$1) \dot{i}_L = -\frac{1}{L} U_{C1} + \frac{1}{L} U$$

$$\dot{U}_{C2} = -\frac{1}{C} i_L + \dot{U}_{C1}$$

$$2) \dot{U}_{C1} = -\frac{1}{RC} U_{C2} + \frac{1}{C} i_L - \frac{1}{RC} U_{C1} + \frac{1}{RC} U$$

$$3) \dot{U}_{C2} = -\frac{1}{RC} U_{C2} - \frac{1}{RC} U_{C1} + \frac{1}{RC} U$$

$$4) Y = U_{C2}$$

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HA) 4)

$$\begin{pmatrix} \dot{U}_{C1} \\ \dot{i}_L \\ \dot{U}_{C2} \end{pmatrix} = \begin{pmatrix} -\frac{1}{RC} & \frac{1}{C} & -\frac{1}{RC} \\ -\frac{1}{L} & 0 & 0 \\ -\frac{1}{RC} & 0 & -\frac{1}{RC} \end{pmatrix} \begin{pmatrix} U_{C1} \\ i_L \\ U_{C2} \end{pmatrix} + \begin{pmatrix} \frac{1}{RC} \\ \frac{1}{L} \\ \frac{1}{RC} \end{pmatrix} U$$

$$Y = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} U_{C1} \\ i_L \\ U_{C2} \end{pmatrix}$$

HA) 1)

$$\Delta \dot{x}(t) = \cancel{\Delta x_2} \begin{pmatrix} \Delta x_2 \\ \frac{2g}{\gamma_R} \Delta x_1 + \left(-\sqrt{\frac{mg}{gc}} L \gamma_R\right) \Delta x_3 \\ \frac{2\sqrt{mgc}}{L \gamma_R} \Delta x_2 + \left(-\frac{R}{L}\right) \Delta x_3 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \frac{1}{L} \Delta b \end{pmatrix}$$

$$\Delta y(t) = 1 \Delta x_1$$

↑

warum ~~State~~ Vektor

wenn $c^T \cdot x$

$\Rightarrow 1 \times 3 \cdot 3 \times 1 \rightarrow 1 \times 1$ "Skalar"

"keine ~~State~~ Änderung
was mit γ_R passiert"